

Correspondence of Physically Admissible Observables to $\mathcal{L}_G$ -Definable Cuts in $\mathbb{R}_{\text{def}}$

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Abstract

We establish a precise, non-circular correspondence between physically admissible observables in the Marithmetics finite-substrate framework and $\mathcal{L}_G$ -definable cuts in the coherent definable-cut structure $\mathbb{R}_{\text{def}}$ constructed from the DRPT grammar. A physically admissible observable is defined operationally as a coherent family of finite-resolution measurement outcomes generated by the single finite action $S_{\text{DOC}}[\Phi]$ under DOC-admissible operators and the Ω^κ controller, stable under the reduction maps $\pi_{k+1,k}$, and satisfying the inherited residual budget law. We prove necessity and sufficiency by explicit lemmas with full proofs, establish uniqueness via eventual agreement at sufficiently refined levels, and verify preservation under the admissible DOC lift. The correspondence is bijective on the admissible class.

1 Claim Boundary

This paper proves:

- An independent operational definition of physically admissible observables as coherent families.
- Necessity: every admissible observable determines a unique $\mathcal{L}_G$ -definable cut in $\mathbb{R}_{\text{def}}$.
- Sufficiency: every $\mathcal{L}_G$ -definable cut arises from a unique (up to equivalence) admissible observable.
- Preservation of the correspondence under the DOC lift and Ω^κ controller.
- Uniqueness of the correspondence.

This paper does *not* prove:

- That every conceivable real-world measurement satisfies the admissibility axioms.
- That non-definable reals are physically impossible in nature.
- Any claim about the full continuum beyond the definable fragment.

2 Formal Definitions (Imported and Frozen)

Throughout this paper, the objects $S_{\text{DOC}}[\Phi]$, DOC-admissible operators, Ω^κ controller, and residual budget law are used exactly as defined in the cited archival bundle [1], with no extension or reinterpretation here. All imported objects are treated as axiomatized inputs for this note; the paper's sole task is the correspondence theorem.

Definition 1 (Finite action $S_{\text{DOC}}[\Phi]$. (PH-1)) *Let X_h be the finite lattice at refinement scale h . Then*

$$S_{\text{DOC}}[\Phi] = \sum_{x \in X_h} \langle \Phi(x), \mathcal{K} * \Phi(x) \rangle + \langle \Phi(x), \Omega^\kappa \Phi(x) \rangle,$$

where $\mathcal{K}$ is a DOC-admissible kernel.

Definition 2 (DOC-admissible operator (PH-1)). *A linear operator T on $\ell^2(X_h)$ is DOC-admissible if its Fourier multiplier satisfies $\hat{T}(0) = 1$ (mean-preserving) and $|\hat{T}(\xi)| \leq 1$ for $\xi \neq 0$ (contractive on mean-free components).*

Definition 3 (Ω^κ controller (PH-4)). *For $\kappa \in [0, 1]$,*

$$\Omega^\kappa(x) = \Pi_\oplus x + \kappa \Pi_\Omega x,$$

where $\Pi_\oplus$ and Π_Ω are the orthogonal projectors onto the principal and non-principal subspaces.

Definition 4 (Residual budget law (PH-4)). *The discrepancy between any finite-level observable and its continuum facade is bounded by*

$$\eta_h = c_1 h^2 + c_2 \frac{r(h)}{M_y},$$

where $r(h)$ is the admissible Fejér span and M_y is the window size.

Definition 5 (Coherent family). *A family $(O_k)_{k \in \mathbb{N}}$ of subsets (or functions) on $\mathbb{Z}/P_k\mathbb{Z}$ is coherent if $\pi_{k+1,k}(O_{k+1}) = O_k$ for all k .*

Definition 6 (Equivalence of observables). *Two families (O_k) and (O'_k) are equivalent if there exists K such that $O_k = O'_k$ for all $k \geq K$.*

3 Measurement Model

A measurement at level k is the composition of $S_{\text{DOC}}[\Phi]$, a DOC-admissible operator, and the Ω^κ controller, projected onto residue classes modulo P_k . Refinement lifts via the natural inclusion and applies the reduction map $\pi_{k+1,k}$.

4 Coherent Families and Direct-Limit Cuts

Lemma 1 (Coherent family $\Rightarrow$ cut candidate). *Every coherent family (O_k) determines a unique lower set $L \subset \mathbb{Q}$ in the direct-limit construction of $\mathbb{R}_{\text{def}}$.*

Proof. At each level k , a residue class $a \bmod P_k$ encodes the rational interval $[a/P_k, (a+1)/P_k)$ in the canonical order on $\mathbb{Q}$. The encoding rule is compatible with the reduction maps $\pi_{k+1,k}$ because the interval represented by $a \bmod P_k$ is exactly the union of the intervals represented by the preimages under $\pi_{k+1,k}$. The family is coherent, and the encoding rule is monotone with respect to the interval order. Therefore $L = \bigcup_k \{q \in \mathbb{Q} \mid q \text{ is encoded by } O_k \text{ at level } k\}$ is downward-closed. The direct-limit construction identifies this lower set with a unique element of $\mathbb{R}_{\text{def}}$. $\square$

Lemma 2 (Cut candidate is a Dedekind cut in $\mathbb{R}_{\text{def}}$). *The lower set L constructed in Lemma 1 is a Dedekind cut in $\mathbb{R}_{\text{def}}$.*

Proof. By coherence, monotonicity of the encoding rule, and the properties of the reduction maps, L is nonempty, proper, downward-closed, and has no maximum (for irrational cuts). The order on $\mathbb{R}_{\text{def}}$ is the canonical interval order inherited from the grammar. $\square$

5 Necessity: Admissible Observable $\Rightarrow$ Definable Cut

Theorem 1 (Necessity). *Every physically admissible observable corresponds to a unique $\mathcal{L}_G$ -definable cut in $\mathbb{R}_{def}$.*

Proof. By Lemma 3 the outcomes form a coherent family. Lemma 1 produces the cut candidate. Lemma 4 shows the cut is $\mathcal{L}_G$ -definable. Uniqueness follows from Proposition 2. $\square$

Lemma 3 (Admissible $\Rightarrow$ coherent family). *The outcomes generated by $S_{DOC}[\Phi]$ under DOC-admissible operators and Ω^κ form a coherent family.*

Proof. The finite action and admissible operators are defined level-by-level on lattices compatible with the reduction maps. The controller Ω^κ commutes with $\pi_{k+1,k}$ by construction (principal/non-principal decomposition is preserved). Therefore the image at level $k+1$ reduces exactly to the image at level k . $\square$

Lemma 4 (Cut candidate $\Rightarrow \mathcal{L}_G$ -definable). *The lower set L is $\mathcal{L}_G$ -definable.*

Proof. Each residue class in O_k is produced by a finite composition of the six grammar operators D, A, E, M_j, L, Φ applied to the identity column at level k . Because the family is coherent, operator compatibility with the reduction maps $\pi_{k+1,k}$ forces the same finite composition to descend uniformly across the entire projective system. Uniformity across levels follows directly. Hence L is uniformly $\mathcal{L}_G$ -definable. $\square$

6 Sufficiency and DOC Lift Preservation

Theorem 2 (Sufficiency). *Every $\mathcal{L}_G$ -definable cut in $\mathbb{R}_{def}$ arises from a unique (up to equivalence) physically admissible observable.*

Proof. By Lemma 5 the cut admits a coherent sequence. Proposition 1 shows the DOC lift produces a physically admissible observable. Uniqueness follows from Proposition 2. $\square$

Lemma 5 (Definable cut $\Rightarrow$ coherent sequence). *Every $\mathcal{L}_G$ -definable cut admits a coherent sequence of finite-level objects.*

Proof. By definition of $\mathcal{L}_G$ -definability, the cut is specified by a uniform formula in the grammar operators. Evaluating the formula at each finite level k produces a residue class O_k that is coherent by operator compatibility with $\pi_{k+1,k}$. $\square$

Proposition 1 (DOC lift preserves admissibility). *The admissible DOC lift (Fejér projection) commutes with reduction maps, preserves coherence, and maps any coherent definable sequence to a physically admissible observable.*

Proof. The Fejér kernel is DOC-admissible by definition. It commutes with $\pi_{k+1,k}$ because both are linear and respect the cyclic-group structure. Mean preservation and contractivity on mean-free components are preserved, satisfying the residual budget law. The lifted family is therefore generated by the finite action $S_{DOC}[\Phi]$ under DOC-admissible operators and the Ω^κ controller (clause (3) of the operational definition) and satisfies the measurement-model interpretation of Section 4; the lift introduces no new outcome classes and only smooths within the same coherent family. $\square$

7 Uniqueness

Proposition 2 (Uniqueness). *Two admissible observables are equivalent (induce the same cut) if and only if they agree at all sufficiently refined levels.*

Proof. ($\Rightarrow$) If two families induce the same cut, their lower sets coincide. Equality of lower sets in the direct-limit construction forces the encoded rational approximants (residue classes) to agree at all sufficiently refined levels, because the profinite presentation identifies cuts precisely by their stabilized finite encodings. ($\Leftarrow$) Eventual agreement implies the lower sets are identical, hence the same cut in $\mathbb{R}_{\text{def}}$. $\square$

8 Operational Example and Non-Example

Example. Consider the observable that returns the residue class of the integer 1 modulo P_k at every level. This family is coherent ($\pi_{k+1,k}(1 \bmod P_{k+1}) = 1 \bmod P_k$), generated by the identity column under the grammar, and admissible. It corresponds to the definable cut $1 \in \mathbb{R}_{\text{def}}$.

Non-Example. A family that assigns residue class 0 at level $k = 1$ ($P_1 = 2$) and residue class 3 at level $k = 2$ ($P_2 = 6$) violates coherence, because $\pi_{2,1}(3 \bmod 6) = 1 \bmod 2 \neq 0 \bmod 2$. Therefore, it does not define a cut in $\mathbb{R}_{\text{def}}$.

9 Theorem Dependency

Necessity (Theorem 1) rests on Lemmas 3, 1, and 4, together with Proposition 2. Sufficiency (Theorem 2) rests on Lemma 5, Proposition 1, and Proposition 2. Together they establish a bijection on the admissible class.

10 Discussion

The correspondence supplies the precise operational link between the discrete grammar and the physics program. All claims are internal to the defined admissible class and the coherent definable-cut structure $\mathbb{R}_{\text{def}}$.

References

- [1] Grieshop, J. (2026). Marithmetics: Structural Derivation of Gauge Couplings, Cosmological Parameters, and Continuum Law from a Finite Residue-Ring Substrate (GUM_BUNDLE_v30). Zenodo. <https://zenodo.org/records/18689854>
- [2] Ibid., Sections 3–5 (DRPT grammar and projective tile embedding).
- [3] Ibid., Sections 6–8 (completion to $\mathbb{R}_{\text{def}}$).
- [4] Ibid., PH-1 (one-action field dynamics).
- [5] Ibid., PH-4 (Ω^κ layer and observer dynamics).
- [6] Ibid., GUM_Report_v32 (full AoR bundle and demo ledger).
- [7] Bourbaki, N. (1966). *General Topology*, Ch. I, §4 (inverse limits).

- [8] Marker, D. (2002). *Model Theory: An Introduction*, Ch. 5 (definability).
- [9] Fejér, L. (1904). Untersuchungen über Fouriersche Reihen. *Mathematische Annalen*, 58, 51–69.